

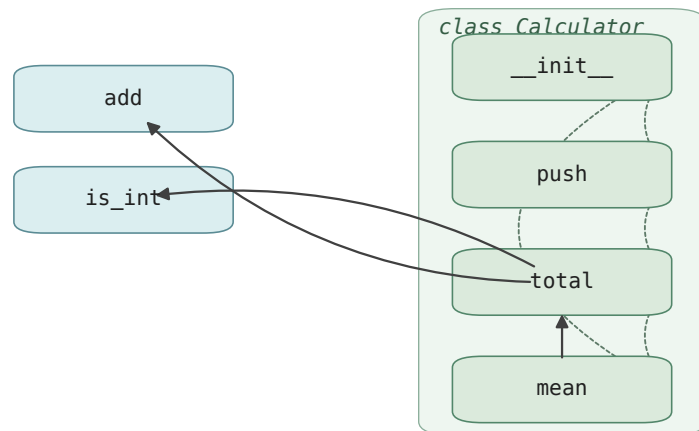
(a) Program dependency graph

```
# calc.py
def add(a, b):
    return a + b

def is_int(x) -> bool:
    """Check if x is an integer."""
    return isinstance(x, int)

class Calculator:
    def __init__(self):
        self.history = []
    def push(self, v):
        self.history.append(v)
    def total(self) -> int:
        """Sum positive integer entries."""
        s, n = 0, 0
        for v in self.history:
            if is_int(v):
                if v > 0:
                    s = add(s, v)
                    n += 1
        return s if n else 0
    def mean(self) -> float:
        return self.total() / max(len(self.history), 1)
```

AST parse



→ $\mathcal{E}_{\text{call}}$

----- $\mathcal{E}_{\text{sib}}$

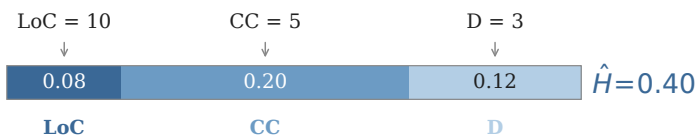
method

top-level

(b) Score breakdown for Calculator.total

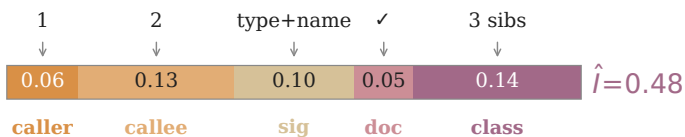
$$\hat{H} = w_l \phi(\text{LoC}, c_l) + w_c \phi(\text{CC}, c_c) + w_d \phi(\text{D}, c_d), \quad \phi(x, c) = \min(x/c, 2)$$

$(w_l, w_c, w_d) = (0.4, 0.4, 0.2), (c_l, c_c, c_d) = (50, 10, 5)$



$$\hat{I} = \alpha C_{\text{caller}} + \beta C_{\text{callee}} + \gamma C_{\text{sig}} + \delta C_{\text{doc}} + \varepsilon C_{\text{class}}$$

$(\alpha, \beta, \gamma, \delta, \varepsilon) = (0.30, 0.25, 0.20, 0.10, 0.15), \text{ each } C_i \in [0, 1]$



$$\text{FIM} = \hat{H} / (\hat{H} + \hat{I} + \varepsilon) = 0.40 \times 0.48 / (0.40 + 0.48) \approx 0.22$$

$\geq \tau = 0.20 \Rightarrow \text{Calculator.total is selected as a mask target.}$

(c) Selection in $\hat{H}$ - $\hat{I}$ plane

